

SHATAYU KULKARNI

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EXPERIENCE

Machine Learning Engineer, Meta Platforms Inc.

New York, NY • Mar. 2022 – Present

- Built end-to-end retrieval and ranking for Dating Crush (Dating referral surface) using one conversion model for retrieval (integrated with Meta's user indexing system) and final ranking; drove +12.5% Dating DAU vs. holdout.
- Replaced the general Dating Home ranking model on People You May Date (PYMD) with a multi-task multi-label transfer-learning model jointly trained on Dating Home and PYMD logs, lifting PRAUC by 0.23 and driving +0.6% Dating DAU.
- Re-architected PYMD ranking infra with a separate cache and request path, enabling P(like)-based candidate ranking and be-spoke model deployment; +1.4% Dating DAU and +0.03% FB App sessions.
- Built a women-only PYMD experience surfacing inbound likers, switching the ranking objective from Dating Home's P(exchange-messages) to PYMD P(like); 2.1x women-initiated likes and +0.77% Dating DAU.
- Built an oncall AI agent for Dating that gathers context on alerts and auto-opens PRs for fixable code issues or supplies re-sources to oncalls (90 engineering hours saved/year, per team survey); led parallel strategy to encode team knowledge in an LLM-accessible format used by all LLM tooling

RESEARCH

Undergraduate Researcher, Purdue CS

West Lafayette, IN • Aug. 2021 – Dec. 2021

- Kulkarni, S. and Azizzadenesheli, K. (2021). *Applying Competitive Policy Gradient to Pokémon Battling* [Undergraduate thesis, Purdue University].
- Applied multi-agent RL algorithm Competitive Policy Gradient (Prajapat et al., 2020) to Pokémon battling via fixed-team self-play; converged to 89–91% winrates against max-damage and random baselines, with the learned agent autonomously discovering non-greedy setup tactics under a sparse ± 1 reward.

Undergraduate Researcher, RCODI Lab

West Lafayette, IN • Jan. 2019 – Oct. 2024

- Kulkarni, S. and Brunswicker, S. (2023, July 17-20). *Maximizing Cooperative Behavior in Collective Action Games Through a Reinforcement Learning AI Agent*. IC2S2 2023 "9th International Conference on Computational Social Science" Copenhagen, Denmark.
- Built a PPO "social planner" that nudges aspiration-based conditional-cooperator agents (Ezaki et al., 2016) into cooperation in iterated public goods games; +14.87% cooperative-contribution proportion, +8.85% total contributions vs. baseline.

INTERNSHIPS

Software Engineer Intern, Meta Platforms Inc.

Menlo Park, CA • Summer 2020 and 2021

- 2021: Developed a feature for easy sharing of interactive charts in code notebooks (500+ uses in first month)
- 2020: Migrated mobile search ad preview in Ads Manager from Hack/PHP to React (part of broader PHP retirement); built comparison + automated testing tooling validating parity across hundreds of thousands of ad previews.

Data Science Intern, Viasat Inc.

West Lafayette, IN • Sept. 2019 – Apr. 2020

- Built ML model for customer Wi-Fi satisfaction; ran clustering analysis on satellite beam performance.

EDUCATION

B.S. in Computer Science (Honors), Purdue University

West Lafayette, IN • Aug. 2018 – Dec. 2021

- GPA: 3.79/4.00
- Minors: Mathematics, Entrepreneurship
- **Selected Coursework:** Reinforcement Learning (graduate level) • Algorithm Design, Analysis, and Implementation (graduate level) • Artificial Intelligence • Data Mining and Machine Learning • Operating Systems • Compilers • Real Analysis

SKILLS

- **Programming Languages:** Python, PyTorch, SQL, C, C++
- **Web Development:** JavaScript, Node.js, React, HTML, CSS, Hack/PHP